

The Course Prologue

Dan Wu

College of Biomedical Engineering and Instrumental Science

danwu.bme@zju.edu.cn

Why image processing?

- What we encounter every day.

Why image processing?



Why image processing?



Why image processing?



Why image processing?



Why image processing?



Why image processing?



Why image processing?

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Google's Nano Banana image model

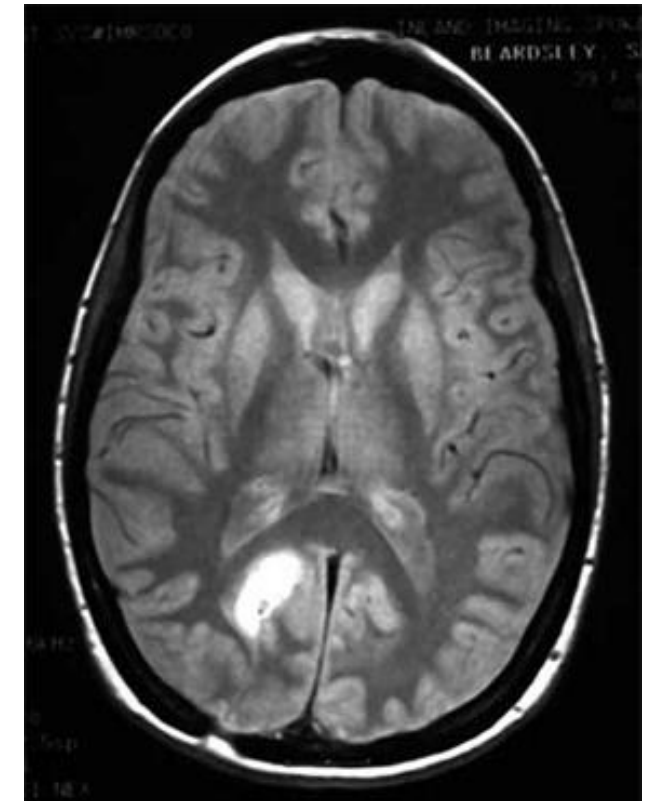
Powered by Google's Gemini 2.5 Flash, Nano Banana turns simple prompts into powerful visuals. Blend references, adjust angles, edit precisely, and maintain perfect character consistency.

Try Nano Banana



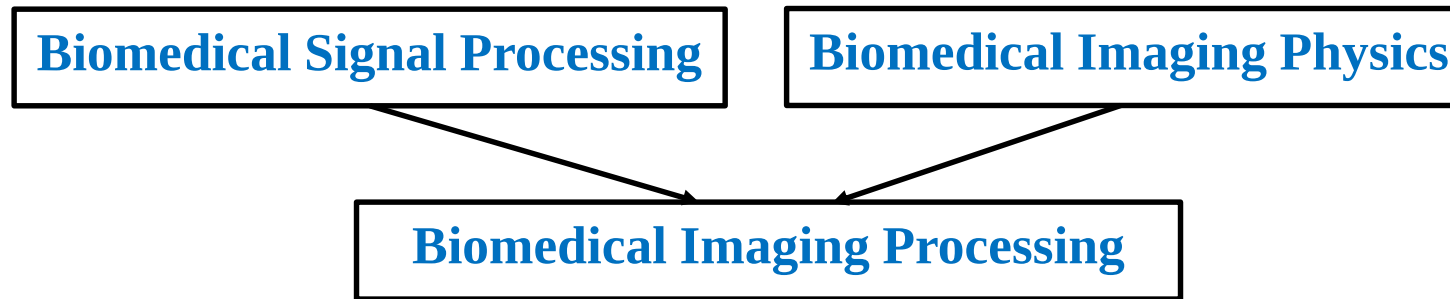
Why Biomedical image processing (BIP)

- When medical doctor received an image
 1. What is this image?
 2. How to make the image looks better?
 3. What's wrong with the patient?
 4. Is this lesion a tumor or what?
 5. How to quantify the lesion?



Why Biomedical image processing (BIP)

- One of major fields in BME.



- Course objectives:
 - Basic concepts of BIP
 - Essential tasks of BIP (filtering, restoration, enhancement, morphological operation, feature extraction, registration, segmentation)
 - Applications in medical imaging

The Contents

时间	授课章次与主要内容	课外学习与作业要求
3.4	BIP绪论 介绍课程安排；基本概念；应用实例；理解视觉感知原理，了解数字图像的生成	课后阅读
3.11	BIP基础 掌握图像的矩阵表示、量化和采样，掌握像素和灰度；了解彩色图像的构成和基本处理思路	实验1
3.18	BIP空间运算 了解图像基本运算、定量测量；空间滤波、卷积操作，卷积核的设计；掌握直方图和灰度变化	作业1
3.25	图像空间增强 掌握几种常用的空间滤波和卷积方法；掌握常用的低通、高通空间滤波器设计；组合滤波设计	实验2
4.1	傅立叶域运算 空间傅立叶变换；离散傅立叶变换特性；快速傅里叶变换；图像小波变换	作业2
4.8	图像频域增强 空间傅立叶的特性；低通与高通频域滤波器	实验3
4.15	图像复原 图像复原处理的基本概念；噪声模型；空间滤波与频域滤波	作业3
4.22	形态学图像处理 形态学处理的基本操作；基本的形态学算法	实验4

Lecture 1-8: Basics of BIP

The Contents

时间	授课章次与主要内容	课外学习与作业要求
4.29	图像分割基础 了解图像分割的定义，边缘连接与边界检测；掌握常用阈值的图像分割方法；区域分割算法	作业4
5.6	图像分割算法 聚类分割；分水岭算法、模版分割算法	实验5
5.13 (调课)	图像配准 医学图像配准定义；图像的变换模型；相似性测度；图像配准中的插值技术	作业5
5.20	图像配准 非线性配准；图像配准方法的评估	实验6
5.27	图像特征提取 边界特征描述；区域特征描述；放射组学	作业6
5.22	医学图像人工智能 机器学习的基本概念；深度机器学习和CNN	实验7
6.3	医学图像人工智能 新技术进展和应用实例	课程大作业
6.10	脑影像处理 常见磁共振脑影像处理算法与流程	课程大作业

Lecture 9-16: Advanced BIP & Medical BIP

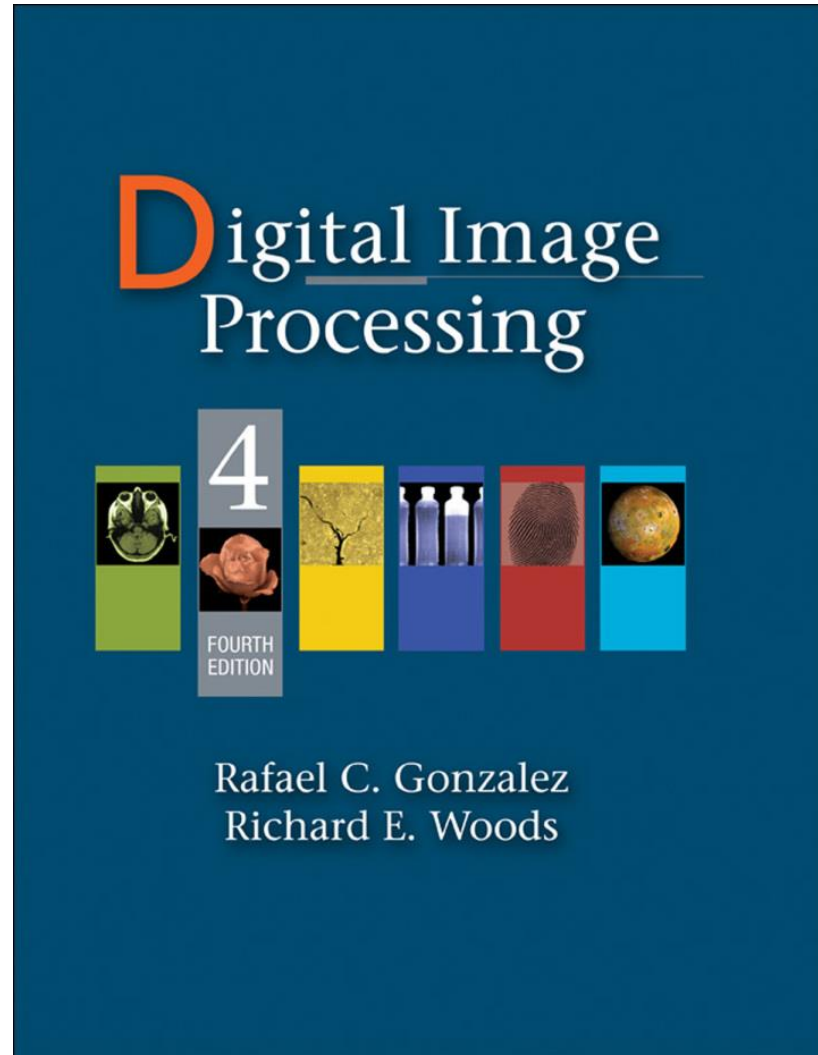
The Course Arrangement

- Course Assignments ($6 \times 3\% = 18\%$)
- Lab Practice ($2\% + 6 \times 3\% = 20\%$)
 - Matlab or Python
- Course Project (12%)
 - Teamwork (3-5), presentation + project reports
 - Around advanced BIP topics
- Final Exam (40%)
- In-class performance (10%)

The Text Book

- Rafael C. Gonzalez and Richard E. Woods. Digital Image Processing (4th Edition), Pearson Education, Inc, 2019
- Rafael C. Gonzalez, Richard E. Woods, and Steven Eddins. Digital Image Processing with Matlab (4rd Edition), Gatesmark Publishing, 2020
- <https://www.imageprocessingplace.com/>
- 吴丹, 何宏建, 张祎, 赵立, 林子暄, 丁秋萍。《磁共振成像原理与应用》, 科学出版社, 2024 (第九、十章)

The Text Book



Rafael C. Gonzalez



Gonzalez is the founder of the Image & Pattern Analysis Laboratory and the Robotics & Computer Vision Laboratory at the University of Tennessee. He also founded Perceptics Corporation

His books are used in more than 1000 educational, industrial, and research institutions in over 50 countries.

MOOC resource

https://mooc1.chaoxing.com/course/portal/ZJLqO4pUUe4jrG-qjIX5Zg==



课程门户 > 链接

生物医学图像处理

AI工作台

任务引擎

课程图谱

班级活动

课件

教案

章节

资料

通知

編輯章節

定時開放任務點提醒 ☐ 導入 導出 打印 批量設置

搜索

目錄

1 基本概念

1 1.1 生物医学图像处理概述

1 1.2 采样与量化

1 1.3 图像基本数学运算

2 图像增强

1 2.1 直方图均衡化与直方图匹配

1 2.2 灰度变换与灰度直方图

1 2.3 频域滤波器设计

默认班级

開放狀態	章節進度
✓ 開放	
✓ 開放	
✓ 開放	
✓ 開放	
✓ 開放	
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✓ 開放	

收起

下載中心

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Related Readings

Journals

- *IEEE Transactions on:*
Image Processing Systems, Man and Cybernetics
Information Theory
Acoustics, Speech and Signal Processing
Pattern Analysis and Machine Intelligence
Medical Imaging
Biomedical Engineering
- ***Medical Image Analysis***
- *Computer Vision and Image Understanding*
- *Artificial Intelligence*
- *Pattern Recognition*
- *Journal of the Optical Society of America*
- *Optical Engineering*
- *Applied Optics-Information Processing*
- *Journal of Electronic Imaging*
- *Proceedings of the IEEE.*

Conferences

- ***AAAI: Conference on Artificial Intelligence***
- ***CVPR: IEEE Conference on Computer Vision and Pattern Recognition***
- ***ICCV: International Conference on Computer Vision***
- ***ICML: International Conference on Machine Learning***
- ***IJCAI: International Joint Conference on Artificial Intelligence***
- ***ECCV: European Conference on Computer Vision***
- ***MICCAI: Medical Image Computing and Computer Aided Intervention***

Other tips

- Course website: course.zju.edu.cn
- Online group: Dingding
 - Lab: 周三第9,10节{双周}, 教六204实验课教室
- Office Hour with teaching assistant: after the lab
- For any questions, feel free to contact
 - Me (danwu.bme@zju.edu.cn)
 - Teaching Assistants: 黄永全 (yongquan.huang@zju.edu.cn) 、
李思荟 (sh.li@zju.edu.cn)

Other tips

- Disclosure for your homework/lab reports/final projects
 - Source code
 - Website/Example
 - Lab resource
 - AI/Copilot
 - Teaching Assistant
 - Which part
 - Other resources
- Let us know your suggestions

